

HMMT Toastober 2026

October 16, 2026

Theam Round

We just finished writing the Theam Round, but a witch came in with a spell and scrambled the problem order! D: Sorry, I guess it's up to you to figure out which problems to tackle.

1. In a haunted house is a machine with a button for each number of the form $2^a \cdot 3^b$ for integers $a \geq 0$ and $b \geq 1$. Each minute, there is a $\frac{1}{21}$ probability that the power fails, after which Ken flees; otherwise he presses exactly one button, and conditioned on the power not failing, the button he presses is the one labeled $2^a \cdot 3^b$ with probability $2^{-a} \cdot 3^{-b}$. All of these choices are independent of everything else. This continues until the power fails. A button is called *cursed* if its label is divisible by 36. Compute the probability that Ken presses at least three (not necessarily distinct) cursed buttons.
2. Given isosceles trapezoid $TOMB$ with $\overline{TO} \parallel \overline{MB}$ and $TO < MB$, rectangle $LIFE$ is inscribed in $TOMB$, with L, I, F , and E on $\overline{TO}, \overline{TB}, \overline{MB}$, and $\overline{OM}$, respectively, such that $\frac{TL}{OL} = 6$, $\frac{TI}{BI} = 4$, and $\frac{MF}{BF} = 5$. Suppose that $LIFE$ has area 56. Compute the area of $TOMB$.
3. Hrishi carves the polynomial 1 onto a pumpkin. Repeatedly, he picks two polynomials A and B on the pumpkin, not necessarily distinct, takes the polynomial

$$x^{17}A(x) + (B(x))^{17},$$

replaces each coefficient with its remainder when divided by 17, and carves it only if it has leading coefficient 1 and degree at most 289. He repeats this until no new polynomial can be carved. Compute the sum of all possible distinct remainders when $P(2)$ is divided by 17 over all polynomials P carved.

4. Let b and o be real numbers such that $b > o > 1$ and

$$\log_b(3o) = \log_o(b^{\log_o b}) = 9.$$

Compute $\log_3(b \cdot o \cdot o)$.

5. For his Halloween party, Daniel cuts out nine paper shapes, each of area 3, and places them flat on a table. Some of the shapes overlap, and the total area of the table covered by the shapes is 22. Let A be the greatest real number with the property that, no matter how the shapes are placed, there exist some two shapes whose intersection has area at least A . Compute A .
6. A coven brews a potion across many nights. Its *potency* is 1 on the first night, 2 on the second night, and for all $n \geq 3$, the product of its potencies on the n th and $(n-1)$ st nights equals its potency on the $(n-2)$ nd night. Compute the number of positive divisors of the potency on the twelfth night.
7. In $\triangle BAT$ with $BA = 7$, $BT = 8$, and $AT = 9$, let M and N be the midpoints of minor arcs $\widehat{AT}$ and $\widehat{AB}$ of the circumcircle of $\triangle BAT$, respectively. Let A' be the reflection of A across the perpendicular bisector of $\overline{BT}$, and let lines $A'M$ and $A'N$ meet line BT at points P and Q , respectively. Let the perpendicular bisector of $\overline{PQ}$ meet the bisector of $\angle BAT$ at H . Compute $AH^2 + A'H^2$.
8. On each four-digit positive integer n , a witch casts a spell such that the number represented by the last three digits of n with any leading zeros removed is divided by 11, and the first digit of n as well as n itself are added to the result. For example, when casted on 2027, the result is $\frac{27}{11} + 2 + 2027 = 2031\frac{5}{11}$, and when casted on 4000, the result is $\frac{0}{11} + 4 + 4000 = 4004$. Find the greatest possible value of n for which the spell casted on n results in an integer palindrome. Recall that a palindrome is a number whose representation reads the same from left to right as from right to left.
9. Call a positive integer n *haunted* if n^n is a perfect sixth power. Find the number of haunted positive integers at most 1000.

10. There are four creatures, a ghost, a goblin, a scarecrow, and a zombie, in a line, where exactly two of the four creatures visited the haunted house. Each creature speaks as follows:

- The ghost: “There is a creature immediately to my left. If that creature visited the haunted house, then I did not visit the haunted house.”
- The goblin: “I did not visit the haunted house.”
- The scarecrow: “If the ghost did not visit the haunted house, then the goblin must have visited the haunted house.”
- The zombie: “There is a creature immediately to my right. If that creature did not visit the haunted house, then I also did not visit the haunted house.”

Suppose that all creatures always tell the truth. Compute the number of possible orderings of the line.